

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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TECHNICAL PROFILE

Embedded firmware engineer developing real-time C for power, control, and communications systems across 16- and 32-bit MCUs, FreeRTOS, and bare-metal DSPs. Builds MATLAB/C++/Python engineering tools, brings up peripherals from hardware documentation, and collaborates with a distributed firmware team on hardware and embedded Linux integration.

TECHNICAL SKILLS

Languages: Embedded C, C++, MATLAB, Python, Bash

Embedded & Real-Time: Bare-metal firmware, FreeRTOS, POSIX threads, interrupts/timers, peripheral/HAL bring-up, boot/update/recovery systems

Processors: dsPIC33CK (16-bit DSC), PIC32, TMS320 DSP, STM32G474, MSP430FR2355; TI AM62 Arm-based SoC

Analysis, Debug & Build: ADC scaling/filter analysis, MATLAB data analysis, CAN-log parsing, JTAG, register inspection, Code Composer Studio, GCC/Make, Git/GitLab, Docker, oscilloscopes, logic/protocol analyzers

Systems & Interfaces: Embedded Linux, Yocto/Arago, device trees, systemd; PWM, ADC, I2C, SPI, UART/RS-232; CAN 2.0, CAN FD, J1939, DroneCAN; IPMI/IPMB

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Develop low-level C firmware for 16-bit dsPIC33CK DSCs and bring up peripherals one subsystem at a time from datasheets, reference manuals, and schematics; isolate register, timing, configuration, and electrical faults using JTAG, targeted tests, oscilloscopes, and logic analyzers.
- Parsed captured CAN output in MATLAB to determine per-channel ADC scaling and compare several filter functions by steady-input variation and time to follow large electrical changes; selected rolling-average and exponential moving average (EMA) approaches and implemented the rolling average in embedded C using a circular buffer and rolling sum.
- Implemented and validated VITA 46.11 Tier 2 management firmware for a VPX power supply, enabling the chassis manager to read IPMI 2.0 sensor, FRU/inventory, and event data over an I2C-based IPMB link.
- Integrate battery hardware and CAN-connected power devices with a TI AM62 Arm-based embedded Linux controller; build Yocto/Arago images and kernels, update device trees, develop external-watchdog services, and trace faults across power, wiring, CAN, boot, configuration, and code.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected work across both internships

- Developed bare-metal C power-control firmware on a TMS320 DSP for transformer-coupled bidirectional buck–boost supplies using 12 PWM A/B pairs (24 physical outputs); configured each A channel's period/frequency, duty cycle, and relative phase while each B output followed or inverted its paired A signal, then validated ADC/PWM behavior in Code Composer Studio.
- Developed embedded C firmware across a two-processor power system: a PIC32 running FreeRTOS coordinated I2C, SPI, CAN/J1939, UART, and update behavior while the bare-metal TMS320 DSP executed real-time converter control.
- Built a resilient update/recovery workflow: a C++ CLI accepted HEX files, a Python TCP sender handled chunking and checksums, and the PIC32 staged new firmware and a golden image in external SPI flash, programmed and verified the DSP over UART, and controlled boot.
- In 2024, built a restricted SSH/chroot interface and Python/Bash commands for a resource-constrained TI Yocto/Arago power system; reverse-engineered RS-232 commands/checksums for programmable loads and relay controllers and helped assemble the associated test circuit.

SELECTED ENGINEERING PROJECTS

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- Developed STM32G474/BQ79616 bring-up and integration for a team-built EV sensor adapter spanning cell-voltage monitoring, ten thermistors sampled by the STM32 ADC, UART, extended-ID CAN telemetry, subsystem-level fault isolation, and custom-PCB hardware.
- Implemented BQ79616 wake/configuration, protocol framing, CRC validation, cell reads, and fault diagnostics; coordinated requirements, staged validation, risk tracking, design reviews, and handoff documentation across the team.

Real-Time Systems & Formal Verification | C, Linux, Lustre

Spring 2026

- Implemented periodic POSIX-thread tasks in C using CPU affinity, absolute-time releases, priority-inheritance mutexes, and timestamp traces; analyzed schedulability, plotted execution timelines, and used bounded model checking on Lustre models to correct temporal monitors and check invariants within configured bounds.

MATLAB Engineering Analysis & Machine Learning

Selected Coursework

- Analyzed measured MOSFET CSV data in MATLAB using numerical gradients and polynomial fits to estimate transconductance, threshold voltage, mobility, and channel-length modulation and compare measured and modeled I–V curves.
- Implemented MATLAB image-learning workflows: trained a CNN to 99.24% validation accuracy, adapted SqueezeNet through transfer learning on a five-class coursework dataset, and applied PCA/KLT to visualize eigenfaces and reduce image dimensionality.

TECHNICAL LEADERSHIP

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- Delivered supplemental lectures and debugged embedded C/MSP430FR2355 firmware and custom-PCB projects across three semesters, diagnosing register, timing, peripheral, toolchain, wiring, and soldering failures across diverse student implementations.